Hospital Patient Analytics

Databricks | PySpark | Delta Lake | AI/BI Dashboards

Independent portfolio project — analytical overview, October 2026

Prepared by Amin Sinaei

Business objective

Explore hospital activity, service-line performance, reported financial outcomes, and patient length of stay. Translate training data into a validated analytical dataset and decision-friendly dashboards that support comparisons across hospitals, service categories, payor segments, regions, and admission priorities.

Dataset and approach

Input: 8,392 service records and 19 original columns. Preparation: inspected schema, checked nulls and complete-row duplicates, validated date ordering and nonnegative financial fields, standardized column names and ZIP codes, and cleaned text fields. Derived fields: length of stay, admission month, and reported profit margin. Output: a Delta analytical table with 8,392 rows and 22 columns.

Dashboard delivery

Created an analytical dashboard covering financial, regional, service, and admission patterns, plus a second interactive Databricks AI/BI dashboard with shared hospital, region, priority, and admission-date filters. Both dashboards use the same prepared analytical data.

Headline results

Total revenue	Reported profit	Distinct patient IDs	Average stay
$18.63M	$1.52M	5,491	2.04 days

Selected findings

North Hospital represented about $9.37M of reported revenue, compared with $7.79M for South Hospital and $1.48M for Downtown Hospital. Surgical and Medical services led revenue at approximately $5.45M and $4.82M respectively. The Low-priority group had an average stay of about 4.24 days, a pattern that warrants further investigation rather than clinical interpretation. Linear correlation between length of stay and revenue was negligible (r approximately 0.0114).

Interpretation and limitations

The source-reported profit could not be reconciled consistently to basic revenue/cost/discount identities; no independent financial validation is claimed. Patient IDs may appear in multiple records, so record count differs from distinct patient count. Length of stay is calculated from admission and discharge dates per record. Findings describe an educational dataset and should not be interpreted as validated hospital benchmarks or medical outcomes.

Dashboard evidence

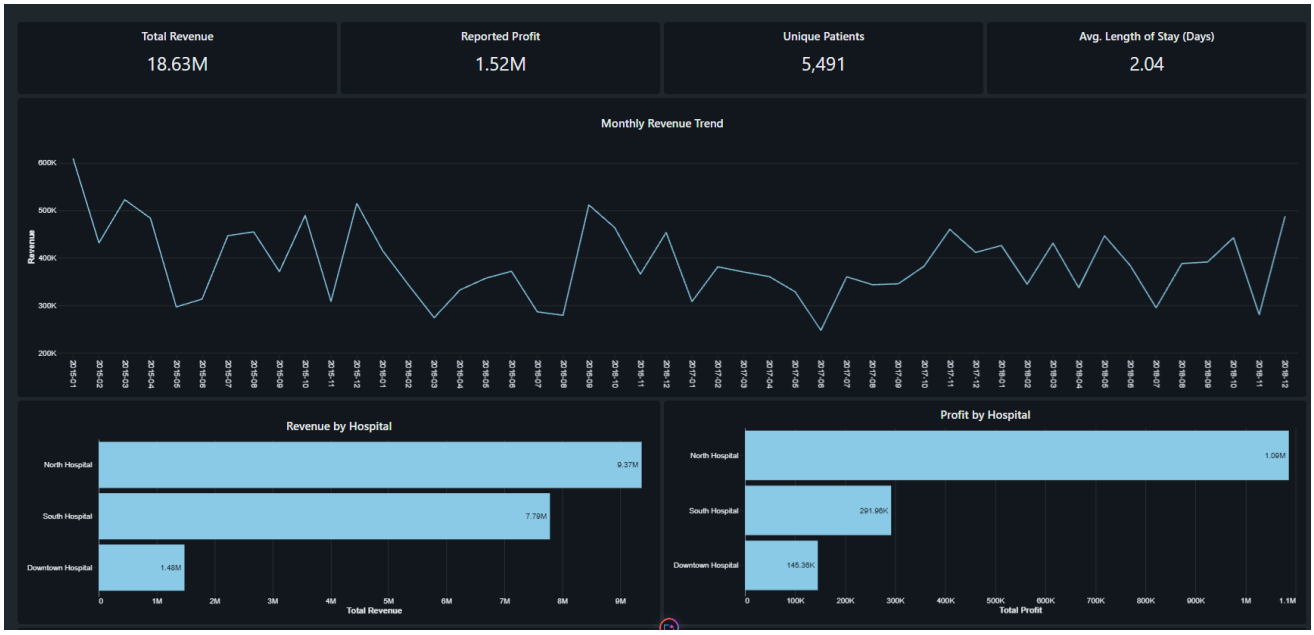


Figure 1. Executive KPIs, monthly revenue, and hospital performance.

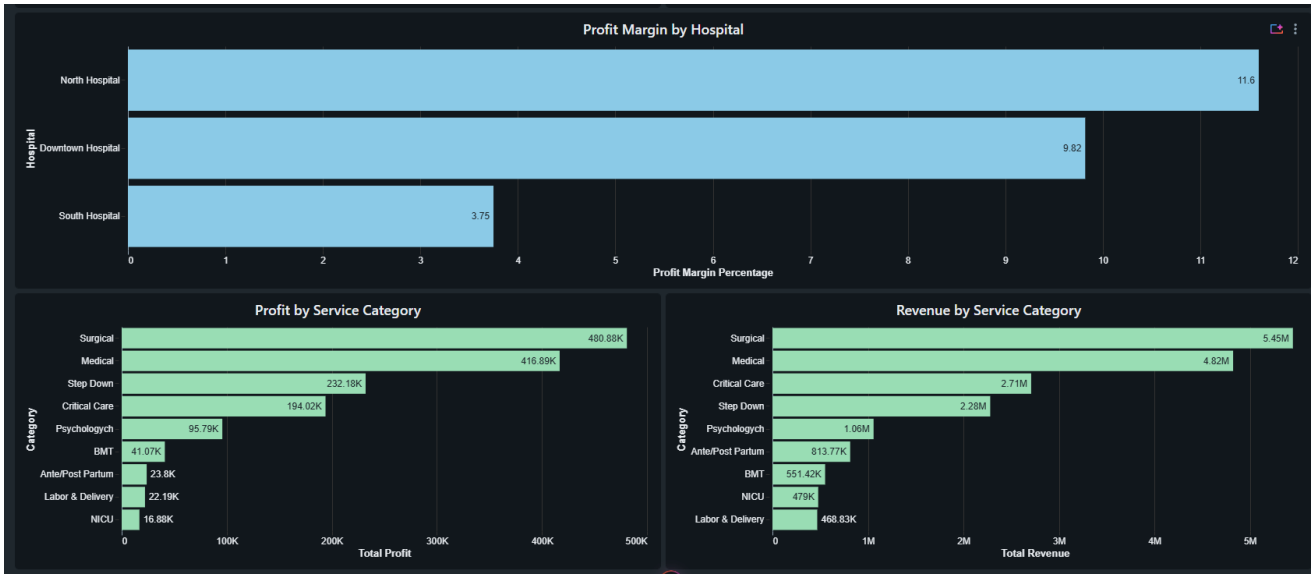


Figure 2. Hospital profit margins and service-category financial comparisons.

Operational and interactive views

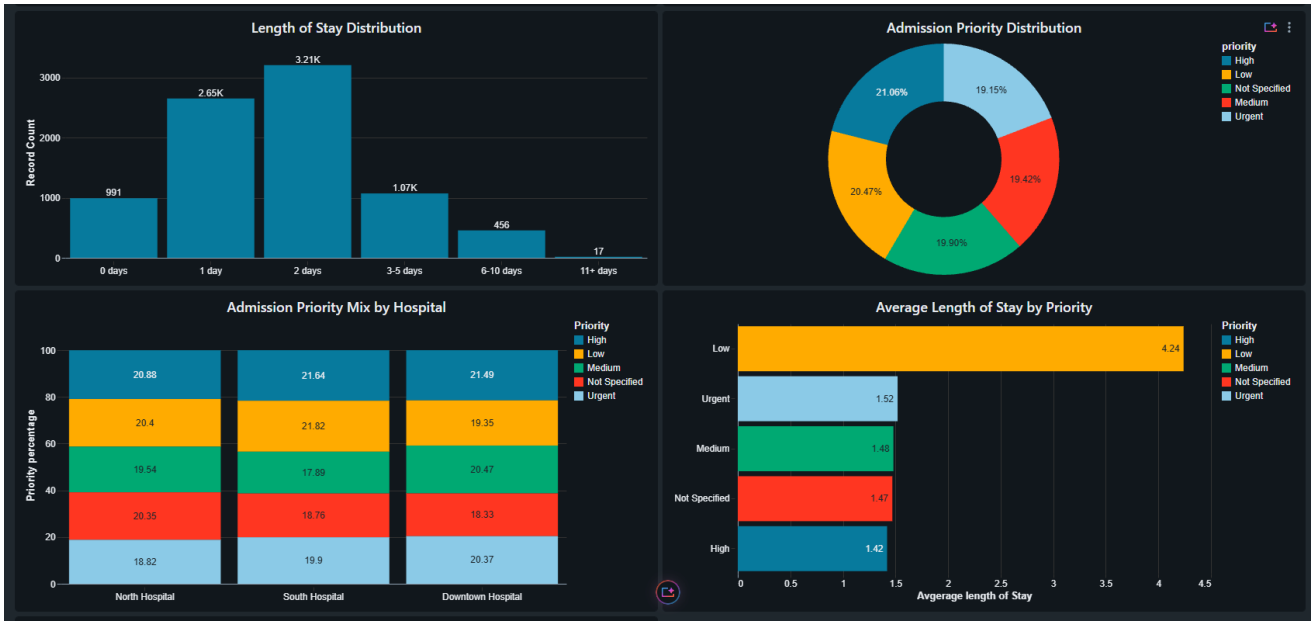


Figure 3. Length-of-stay distribution and admission-priority comparisons.

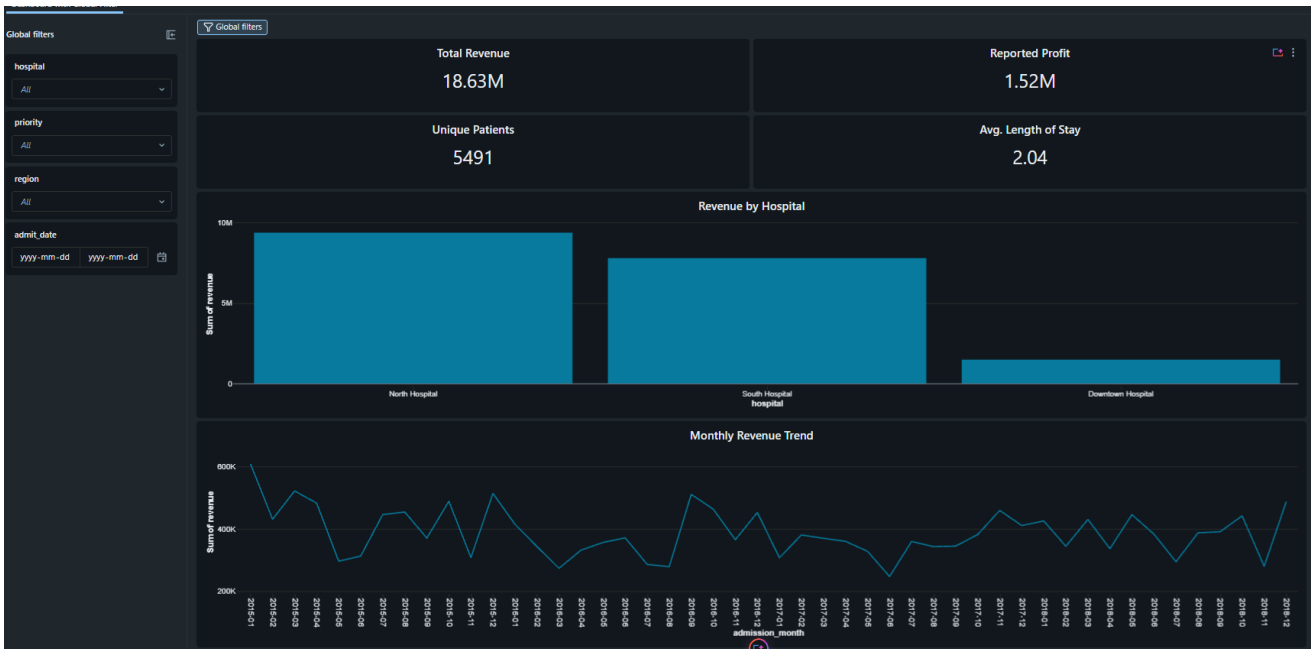


Figure 4. Interactive dashboard with shared hospital, region, priority, and date filters.

This document is a portfolio-level overview. It intentionally omits line-by-line SQL and does not claim to provide an executable project reproduction.